Week 7 – Quantum Computing

Eigenstuff, Grover's Algorithm

Vikram Kamat

Linear Algebra: Revisiting Eigenvalues and Eigenvectors

Before proceeding with our study of quantum algorithms, we pause to review a concept from linear algebra that is central throughout quantum mechanics and quantum computing: *eigenvalues* and *eigenvectors*.

Eigenvectors: invariant directions

Let A be a linear operator on a vector space. A nonzero vector $|v\rangle$ is called an *eigenvector* of A if

$$A|v\rangle = \lambda|v\rangle$$

for some scalar $\lambda \in \mathbb{C}$, called the *eigenvalue*.

Geometrically, this means that $|v\rangle$ is a direction that is preserved by A: the operator may stretch it or rotate its phase, but it does not move it to a different direction in the vector space.

Example. For the Pauli matrix $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, we have $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$. Thus $|0\rangle$ and $|1\rangle$ are eigenvectors with eigenvalues $+1$ and -1, respectively.

Measurement and eigenstates

Recall that measuring a qubit in the computational basis produces outcomes corresponding to $|0\rangle$ or $|1\rangle$. In *eigen language*, this is stated as: *measuring in the computational basis is equivalent to measuring the observable Z.*

The possible measurement outcomes are the eigenvalues of Z, namely $+1$ and -1, and the post-measurement states are the corresponding eigenvectors $|0\rangle$ and $|1\rangle$.

More generally, in quantum mechanics, **measurement outcomes correspond to eigenvalues of *Hermitian* operators, and the system collapses into the associated eigenstate.**

Eigenbases

For a linear operator A on a finite-dimensional vector space V, a *set of eigenvectors* $\{|v_1\rangle, \ldots, |v_n\rangle\}$ is called an *eigenbasis* of A if:

- each $|v_i\rangle$ is an eigenvector of A, and

- the set $\{|v_1\rangle, \ldots, |v_n\rangle\}$ forms a basis for V.

In this case, every vector in V can be written as a linear combination of eigenvectors of A. With respect to such a basis, the operator A acts in a particularly simple way: it merely rescales each basis vector by its corresponding eigenvalue.

Example. As seen above, the Pauli matrix Z has eigenbasis $\{|0\rangle, |1\rangle\}$, while the Pauli-X matrix (as you will show in the exercise below) has eigenbasis $\{|+\rangle, |-\rangle\}$.

Note. This perspective also clarifies the BB84 protocol studied earlier: choosing between the computational basis $\{|0\rangle, |1\rangle\}$ and the $\{|+\rangle, |-\rangle\}$ basis corresponds to measuring in the eigenbases of Z and X, respectively. The security of BB84 relies on the fact that these observables have different eigenbases.

Another note. Not every linear operator admits an eigenbasis; however, Hermitian and unitary operators on finite-dimensional complex vector spaces always admit an orthonormal eigenbasis.

Exercise 1.

(a) Show that $|0\rangle$ and $|1\rangle$ are eigenvectors of the Pauli-Z matrix. What are the corresponding eigenvalues?

(b) Show that $|+\rangle = \frac{|0\rangle+|1\rangle}{\sqrt{2}}$ and $|-\rangle = \frac{|0\rangle-|1\rangle}{\sqrt{2}}$ are eigenvectors of the Pauli-X matrix. What are the corresponding eigenvalues?

Exercise 2. *Let $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ be a qubit state.*

(a) Express $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis.

(b) If we measure $|\psi\rangle$ in the $\{|+\rangle, |-\rangle\}$ basis, what are the probabilities of each outcome?

(c) Verify that measuring in the $\{|+\rangle, |-\rangle\}$ basis is operationally equivalent to applying a Hadamard gate followed by a computational basis measurement, i.e. the measurement probability for $|+\rangle$ in the first case is identical to the measurement probability for $|0\rangle$ in the second case (and similarly for $|-\rangle$ and $|1\rangle$).

Exercise 3. *Let H be a Hermitian operator and suppose $H|v\rangle = \lambda|v\rangle$ for some nonzero vector $|v\rangle$.*

(a) Compute $\langle v|H|v\rangle$ in two different ways: first using $H|v\rangle = \lambda|v\rangle$, and second using the fact that $H = H^\dagger$.

(b) Conclude that λ must be real.

Grover search: a query-model formulation

The centerpiece of this (and the next) lecture involves the study of the algorithmic problem of **unstructured search**, which, as with the Deutsch–Jozsa (and Bernstein–Vazirani) algorithms, is best expressed within the *query model of computation*, with an additional **promise**. We are given black-box (oracle) access to a Boolean function

$$f : \{0,1\}^n \to \{0,1\},$$

and we wish to *learn* about f using as few queries to this oracle as possible.

The unstructured search problem

To model unstructured search, we define f to be an **indicator** function of some hidden subset

$$S = \{x \in \{0,1\}^n : f(x) = 1\}.$$

The **search task**, then, is to find an element of S, i.e. to output some $x^\star \in \{0,1\}^n$ such that $f(x^\star) = 1$ (if such an element exists).

The promise – a unique solution. To initially simplify matters, and to keep the analysis clean, we assume the promise $|S| = 1$, i.e. there exists a unique *marked item* $x^\star$ such that $f(x^\star) = 1$, and $f(x) = 0$ for all $x \neq x^\star$.

Oracle models: O_f and Z_f

We've seen as part of the Deutsch–Jozsa algorithm that a Boolean function $f : \{0,1\}^n \to \{0,1\}$ is implemented as the following unitary *oracle*:

$$O_f : |x\rangle\,|y\rangle \longmapsto |x\rangle\,|y \oplus f(x)\rangle\,.$$

For the analysis of Grover's algorithm, it will be more convenient to work with an equivalent *phase oracle* given by

$$Z_f : |x\rangle \longmapsto (-1)^{f(x)}\,|x\rangle\,.$$

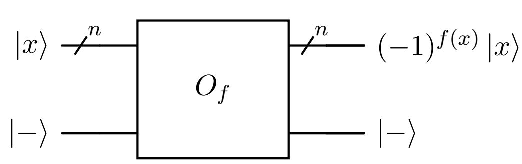
From O_f to Z_f via phase kickback. If we initialize the workspace qubit $|y\rangle = |-\rangle$, we've seen during analysis of the Deutsch–Jozsa algorithm that applying O_f to $|x\rangle\,|-\rangle$ yields

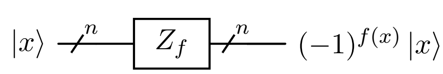
$$O_f\big(|x\rangle\,|-\rangle\big) = (-1)^{f(x)}\,|x\rangle\,|-\rangle\,.$$

Thus, on the first register the effect is precisely that of the phase oracle Z_f:

$$|x\rangle \longmapsto (-1)^{f(x)}\,|x\rangle\,.$$

Thus, we assume that Z_f is O_f with a fixed workspace qubit $|-\rangle$.





We are now ready to formally define the search problem that Grover's algorithm tackles, using a technique called *quantum amplitude amplification*.

Definition 1. *Given query access to an oracle Z_f for an unknown Boolean function $f : \{0,1\}^n \to \{0,1\}$ with the promise that there exists a unique marked item $x^\star \in \{0,1\}^n$ such that $f(x^\star) = 1$, output $x^\star$ with high probability using as few oracle queries as possible.*

Before we dive into the formal details of the complete algorithm, it might be instructive to see the *amplitude amplification* idea on a small example.

An example on 4 qubits

Let $n = 4$, so the search space has size $N = 2^4 = 16$. We assume that Z_f is the phase oracle encoding the function $f : \{0,1\}^4 \to \{0,1\}$ that "marks" the input string $x^\star = 1101$. In other words, $f(x^\star) = 1$, while $f(x) = 0$ for each $x \neq x^\star$.

Step 0: Uniform superposition

Given that the algorithm does not know which of the inputs is marked, it begins with the following uniform superposition over all 4 bit strings:

$$|s\rangle = |\psi_0\rangle = \frac{1}{\sqrt{16}} \sum_{x \in \{0,1\}^4} |x\rangle = \begin{pmatrix} 1/4 \\ 1/4 \\ \vdots \\ 1/4 \end{pmatrix}.$$

Every basis state has amplitude $\frac{1}{4}$, and hence probability $\left|\frac{1}{4}\right|^2 = \frac{1}{16}$.

Step 1: Phase oracle

Let Z_f denote the phase oracle with action $Z_f |x\rangle = (-1)^{f(x)} |x\rangle$. Since $f(x^\star) = 1$ and $f(x) = 0$ otherwise, applying Z_f produces

$$|\psi_1\rangle = \begin{pmatrix} 1/4 \\ 1/4 \\ \vdots \\ -1/4 \\ \vdots \\ 1/4 \end{pmatrix},$$

where only the entry corresponding to $x^\star = 1101$ has changed sign.

Note: It is useful that the oracle has "marked" exactly the string we are searching for, and no others. However, by itself it isn't sufficient since the probability of measuring $x^\star$ remains $\left|-\frac{1}{4}\right|^2 = \frac{1}{16}$, exactly the same as any other (unmarked) input string.

Step 2: Inversion about the mean

This step is key. In short, what the algorithm will do is "amplify" exactly the marked amplitude by carrying out an operation that amounts to inverting about the mean.

For $N = 2^n$, let $J = J_N$ denote the $N \times N$ matrix of all ones, and define $M = \frac{1}{N} J$. It is clear to see that for any n-qubit state $|v\rangle$, $M|v\rangle$ replaces each entry of $|v\rangle$ with the average of the entries of $|v\rangle$.

Exercise 4 (Diffusion operator)**.** *Show that the operator $D = 2M - I$ is unitary.*

The Grover iterate

The Grover iterate is defined as $G = DZ_f$, i.e. apply the phase oracle Z_f, followed by the diffusion operator D. Let's apply it to our specific 4-qubit example, where $N = 16$.

After the phase oracle, we had:

$$15 \text{ entries equal to } \frac{1}{4}, \qquad 1 \text{ entry equal to } -\frac{1}{4}.$$

The average amplitude is therefore

$$\frac{15(\frac{1}{4}) + (-\frac{1}{4})}{16} = \frac{14}{16} \cdot \frac{1}{4} = \frac{14}{64} = \frac{7}{32}.$$

Applying $D = 2M - I$ reflects each amplitude about this average. For the unmarked entries:

$$2\left(\frac{7}{32}\right) - \frac{1}{4} = \frac{14}{32} - \frac{8}{32} = \frac{6}{32} = \frac{3}{16}.$$

For the marked entry:

$$2\left(\frac{7}{32}\right) - \left(-\frac{1}{4}\right) = \frac{14}{32} + \frac{8}{32} = \frac{22}{32} = \frac{11}{16}.$$

Thus after one Grover iteration:

$$|\psi_2\rangle = \begin{pmatrix} \frac{3}{16} \\ \vdots \\ \frac{11}{16} \\ \vdots \\ \frac{3}{16} \end{pmatrix}.$$

The probability of the marked element is now $\left(\frac{11}{16}\right)^2 = \frac{121}{256} \approx 0.473$, an improvement over $\frac{1}{16}$.

Exercise 5.

1. *Apply a second and a third Grover iteration to $|\psi_2\rangle$ and compute the respective probabilities of measuring the marked input string.*
2. *Now apply a fourth Grover iteration to the state resulting from the third iteration. What do you notice?*

Analysis of Grover's Algorithm: A Geometric lens

The effect of both the phase oracle Z_f and the diffusion operator $D = 2M - I$ on the superposition state admits an elegant geometric interpretation of the Grover iterate that we use to analyze the algorithm in this section.

Suppose $N = 2^n$, and let $|x^\star\rangle$ be the marked basis state. Let $|s\rangle = \frac{1}{\sqrt{N}} \sum_{x \in \{0,1\}^n} |x\rangle$ be the uniform superposition over all N basis states. As seen in the example above, Grover's algorithm works as follows:

1. Begin with $|s\rangle$.
2. Apply the Grover iterate $G = DZ_f$ a total of t times, for some optimal value of t that will be determined soon.

To begin our analysis of the algorithm, let

$$|g\rangle = |x^\star\rangle\,, \qquad |b\rangle = \frac{1}{\sqrt{N-1}} \sum_{x \neq x^\star} |x\rangle\,.$$

Informally, $|g\rangle$ is the "good" marked state, and $|b\rangle$ is the uniform superposition over all "bad" unmarked states. Verify that vectors $|g\rangle$ and $|b\rangle$ are orthonormal; furthermore, verify also that the uniform superposition may be written as

$$|s\rangle = \frac{1}{\sqrt{N}}\,|g\rangle + \sqrt{\frac{N-1}{N}}\,|b\rangle\,.$$

What this tells us is that $|s\rangle$ lies in the 2-dimensional subspace spanned by $|g\rangle$ and $|b\rangle$. Both Z_f and D preserve this property, i.e. the algorithm evolves **entirely within this subspace**.

The following exercises will help us learn about the *reflection* properties that both Z_f and D have.

Exercise 6.

1. *Show that the outer product* $|s\rangle\langle s| = \frac{1}{N} J_N$*, where* J *is the* $N \times N$ *all* 1*'s matrix.*
2. *Show that for the diffusion operator given by* $D = 2\,|s\rangle\langle s| - I$*,* $D\,|s\rangle = |s\rangle$*, and* $D\,|s^\perp\rangle = -\,|s^\perp\rangle$ *for any* $|s^\perp\rangle$ *that is orthogonal to* $|s\rangle$*.*
3. *Show that* $Z_f\,|g\rangle = -\,|g\rangle$ *and* $Z_f\,|b\rangle = |b\rangle$*.*

Corollary. Exercises 2 and 3 imply that the operators D and Z_f reflect a state about $|s\rangle$ and $|b\rangle$ respectively. Figure 1 demonstrates one iteration of the composition DZ_f: rotation by 2θ in the direction of $|g\rangle$.

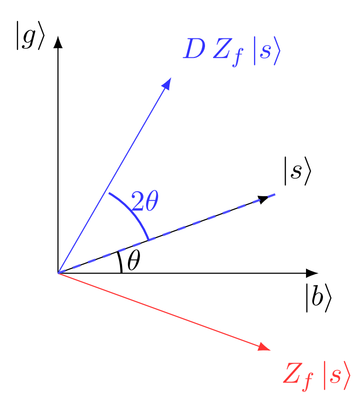


Figure 1: Z_f reflects about $|b\rangle$, while the diffusion operator $D = 2\,|s\rangle\langle s| - I$ reflects about $|s\rangle$. If θ is the angle between $|s\rangle$ and $|b\rangle$, the net effect in the $\{|g\rangle\,, |b\rangle\}$ plane is a rotation by 2θ toward $|g\rangle$.

Number of iterations

The inner product $\langle s|b\rangle = \sqrt{\dfrac{N-1}{N}}$. Referring to Figure 1, this allows us to write $\cos(\theta) = \sqrt{\dfrac{N-1}{N}}$, and $\sin(\theta) = \sqrt{\dfrac{1}{N}}$.

Then the initial state can be written as

$$|s\rangle = \sin\theta\ |g\rangle + \cos\theta\ |b\rangle\,.$$

As application of the Grover iterate $G = DZ_f$ rotates the state by angle 2θ toward $|g\rangle$, the state after k iterations can be expressed as:

$$G^k\,|s\rangle = \sin\big((2k+1)\theta\big)\,|g\rangle + \cos\big((2k+1)\theta\big)\,|b\rangle\,.$$

After k iterations, this probability of measuring $|g\rangle$ is $\sin^2\big((2k+1)\theta\big)$. To maximize this probability – i.e. make it as close to 1 as possible, we want $(2k+1)\theta \approx \frac{\pi}{2}$. Solving for k yields $k \approx \frac{\pi}{4\theta} - \frac{1}{2}$.

Since $\sin\theta = \frac{1}{\sqrt{N}}$, we have $\theta = \sin^{-1}\left(\frac{1}{\sqrt{N}}\right)$. When N is large, we have $\theta = \sin^{-1}\left(\frac{1}{\sqrt{N}}\right) \approx \frac{1}{\sqrt{N}}$ since $sin(\theta) \approx \theta$ for small θ. Substituting into the equation for k, we get

$$k \approx \frac{\pi}{4}\sqrt{N} - \frac{1}{2}.$$

In other words, repeating the Grover iterate approximately $k \approx \frac{\pi}{4}\sqrt{N}$ times concentrates nearly all amplitude on the marked state.

Overshooting the mark. It is important to note that Grover's algorithm does not "amplify" the marked amplitude monotonically. To understand this geometrically, we refer again to Figure 1. Each Grover iterate rotates the state vector by a fixed angle 2θ in the $\{|g\rangle, |b\rangle\}$ plane. The success probability is largest when the state is closest to the $|g\rangle$ axis. If we apply too many iterations, the state rotates *past* the $|g\rangle$ direction (into the second quadrant) and begins moving away from $|g\rangle$ again, causing the success probability to decrease. Hence the importance of choosing k carefully: too few leaves significant amplitude on $|b\rangle$, while too many rotates amplitude back toward $|b\rangle$.

Concluding remarks: Quadratic speedup

Classically, since the function is a black box with no structure to exploit, each query rules out at most one candidate, so in the worst case you require $\Theta(N)$ queries to find the marked element.

Thus, Grover's algorithm provides a **quadratic** speedup by requiring $\Theta(\sqrt{N})$ queries. [1]

[1] The *Theta* notation $\Theta(g(n))$ provides an asymptotically tight bound on a function, indicating that the algorithm's running time grows at the same rate as the function $g(n)$ for large inputs.